

This document outlines the core user interface wireframes for KGP-Assist, an AI-powered assistant platform at IIT Kharagpur. Each screen is designed following established UX principles to ensure clarity, usability, and accessibility across web and mobile platforms.

1. Dashboard / Home Screen

Design Rationale

This screen aligns with user-centered design principles outlined in Garrett’s Five Planes (Structure & Skeleton), Krug’s call for clarity and simplicity, and Buxton’s early ideation approach. Visual hierarchy and intuitive layout support efficient interaction.

Stakeholder Impact

Improves administrative efficiency and reduces query load by guiding users clearly through key processes. Supports data-driven optimization by capturing interaction data and reducing reliance on fragmented systems.

Wireframe Placeholder

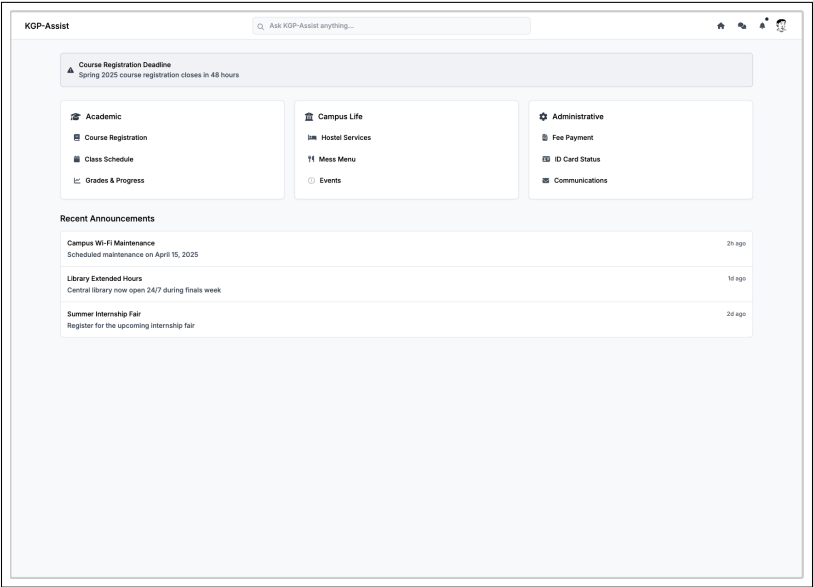


Figure 1: Placeholder for wireframe sketch of this screen.

Purpose

Provide a personalized entry point with role-specific actions and announcements.

Key UI Components

- Role-based dashboard cards (Student, Faculty, Staff)
- Global navigation bar with Home, Chat, Notifications, and Profile
- Prominent search box for “Ask KGP-Assist”
- Announcements and urgent alerts

Intended Actions

- Select a task card (e.g. course registration)
- Interact with chatbot or type queries
- Navigate to detailed views or alerts

Responsiveness and Accessibility

- Responsive grid for mobile stacking
- High-contrast design and ARIA labeling

2. Conversational Interface

Design Rationale

This screen aligns with user-centered design principles outlined in Garrett's Five Planes (Structure & Skeleton), Krug's call for clarity and simplicity, and Buxton's early ideation approach. Visual hierarchy and intuitive layout support efficient interaction.

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Wireframe Placeholder

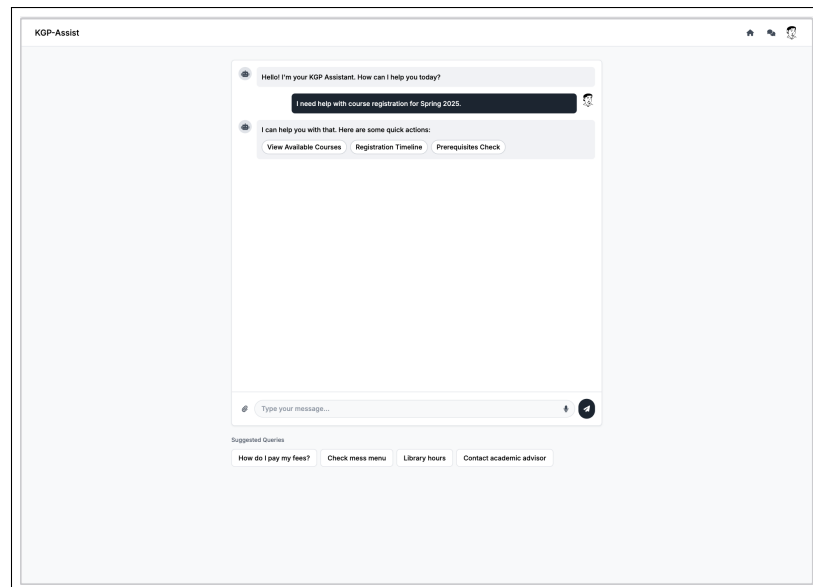


Figure 2: Placeholder for wireframe sketch of this screen.

Purpose

Enable natural language interaction for queries and requests using chat or voice.

Key UI Components

- Chat panel with clear user/assistant messages
- Text + voice input with send button
- Quick action suggestions and upload support

Intended Actions

- Enter query or use voice
- Select from suggested intents
- Upload forms or request help

Responsiveness and Accessibility

- Chat interface scales to mobile
- Live region updates for screen readers

3. Course Registration Flow

Design Rationale

This screen aligns with user-centered design principles outlined in Garrett's Five Planes (Structure & Skeleton), Krug's call for clarity and simplicity, and Buxton's early ideation approach. Visual hierarchy and intuitive layout support efficient interaction.

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Wireframe Placeholder

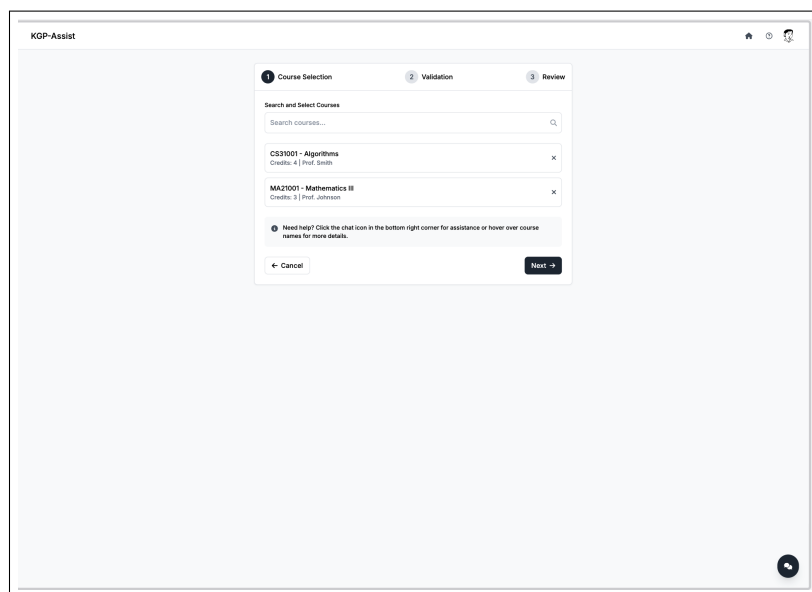


Figure 3: Placeholder for wireframe sketch of this screen.

Purpose

Guide students through registration in 3 simple steps using validation.

Steps and UI

1. Course selection: searchable dropdowns
2. Rule validation and warning display
3. Final review and submit screen

Additional Features

- Progress indicator and Back/Next buttons
- Contextual help via chat and info tooltips

4. Hostel Allocation Flow

Design Rationale

This screen aligns with user-centered design principles outlined in Garrett's Five Planes (Structure & Skeleton), Krug's call for clarity and simplicity, and Buxton's early ideation approach. Visual hierarchy and intuitive layout support efficient interaction.

Stakeholder Impact

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Wireframe Placeholder

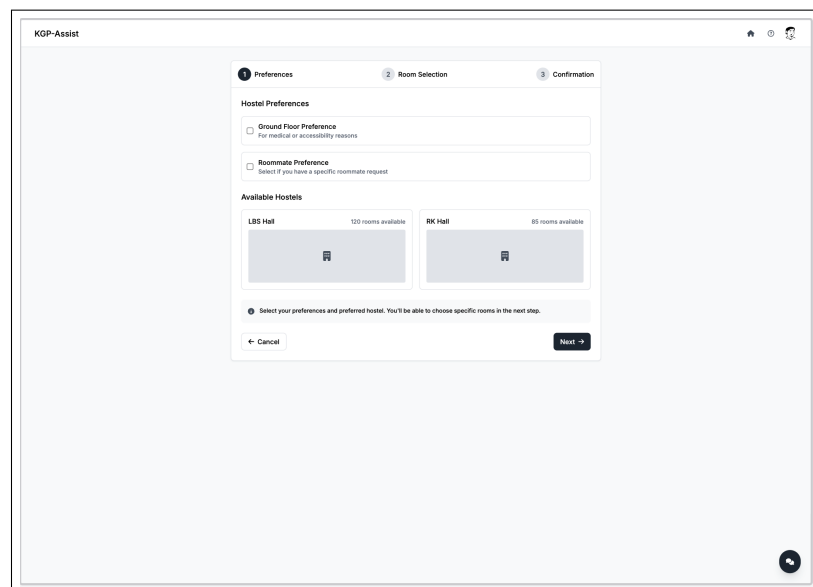


Figure 4: Placeholder for wireframe sketch of this screen.

Purpose

Allow students to submit preferences and select hostel rooms.

UI and Actions

- Step-by-step wizard with room selection map/list
- Preference checklist and final confirmation

5. Issue Tracking

Design Rationale

This screen aligns with user-centered design principles outlined in Garrett's Five Planes (Structure & Skeleton), Krug's call for clarity and simplicity, and Buxton's early ideation approach. Visual hierarchy and intuitive layout support efficient interaction.

Stakeholder Impact

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Wireframe Placeholder

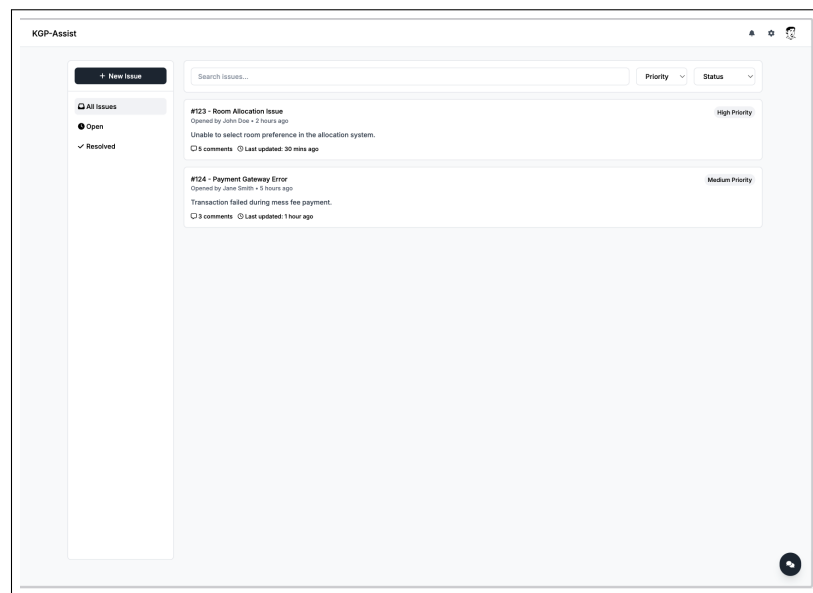


Figure 5: Placeholder for wireframe sketch of this screen.

Purpose

Track support queries and tickets raised with real-time updates.

UI Components

- Ticket list with filters and status
- Detail view with activity log and comments
- Report New Issue button

6. Document Submission & Form Assistance

Design Rationale

This screen aligns with user-centered design principles outlined in Garrett's Five Planes (Structure & Skeleton), Krug's call for clarity and simplicity, and Buxton's early ideation approach. Visual hierarchy and intuitive layout support efficient interaction.

Stakeholder Impact

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Wireframe Placeholder

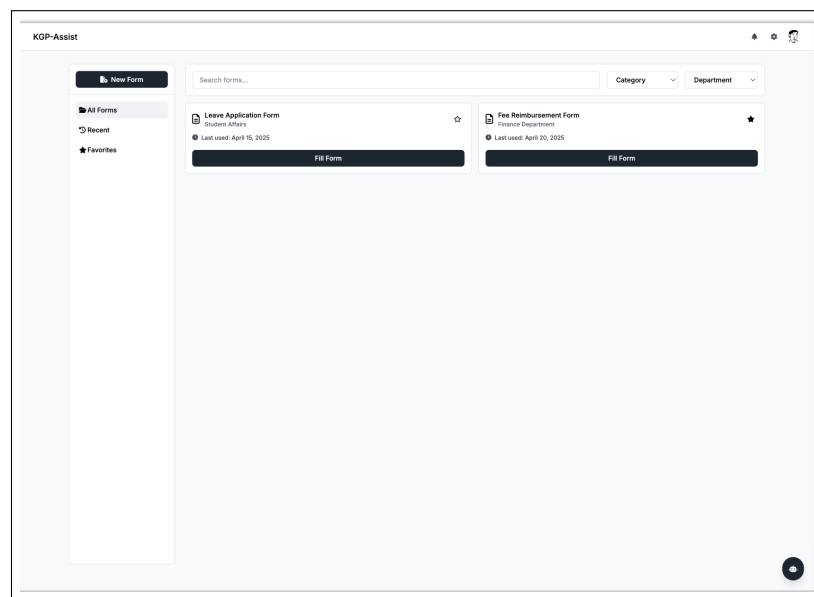


Figure 6: Placeholder for wireframe sketch of this screen.

Purpose

Help users fill and submit institutional forms with AI assistance.

Features

- Form library and search
- Auto-filled forms with preview and validation
- Upload interface for supporting files

7. Integrated Campus Systems

Design Rationale

This screen aligns with user-centered design principles outlined in Garrett's Five Planes (Structure & Skeleton), Krug's call for clarity and simplicity, and Buxton's early ideation approach. Visual hierarchy and intuitive layout support efficient interaction.

Stakeholder Impact

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Wireframe Placeholder

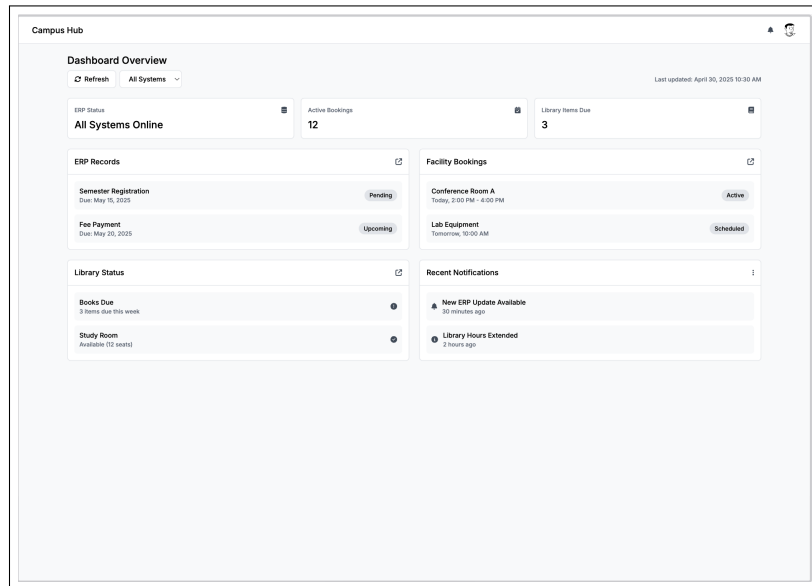


Figure 7: Placeholder for wireframe sketch of this screen.

Purpose

Surface ERP, booking, and library info in one dashboard.

Design Notes

- Data cards showing ERP records or facility calendars
- Manual refresh, system source labels, and direct links